Subject: Bio301 (Notes)

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1). Which of following are obtained from fruits, vegetables and cereals?

A. Monosaccharide's
B. Sucrose
C. Cellulose
D. Starch

2) Molecular formula of fructose is

A. C12H22O11
B. C18H32O16
C. C6H12O6
D. None of these

3) pon hydrolysis oligosaccharides form

A. 1 to 8 molecules of simple sugars
B. 10 to 12 molecules of simple sugars
C. 5 to 8 molecules of simple sugars
D. 2 to 9 molecules of simple sugars

4) Starch is an example of?

A. Monosaccharides
B. Oligosaccharides
C. Polysaccharides
D. Lipids

5) When glucose molecules are joined by α 1-4 linkage, type of carbohydrate obtained is

A. lactose
B. starch
C. glycogen
D. Lipids

6) When mixed with iodine glycogen turns

A. blue
B. purple
C. pink
D. red

7) Glucose is a monosaccharide and is a

A. hexose
B. pentose

C. furanose
D. sucrose

8) Simplest form of sugars are usually

A. colorless
B. water soluble
C. crystalline
D. all of above

9) When 2 to 20 monosaccharidic units are linked by glycosidic bonds than they form

A. polysaccharides
B. monosaccharides
C. oligosaccharides
D. Disaccharides

10) Suffix used in carbohydrate naming is

A. number of hydrogen atoms
B. number of carbon atoms
C. number of carbonyl groups
D. D or L configuration

11) Difference in disaccharides is due to

A. number of H atoms
B. arrangement of atoms within
C. number of C atoms
D. number of O atoms

12) Hydrolysis of glycoside bond involves

A. breakdown of glycosidic bonds
B. formation of glycosidic bonds
C. formation of hydrogen bond
D. formation of ionic bond

13) In naming cyclic carbohydrates anomers are placed in the

A. beginning
B. end
C. middle
D. termination

14) When all monosaccharides in a polysaccharide are same type than polysaccharide is called

A. oligosaccharide
B. glycogen
C. homoglycan
D. Heteroglycans

15) Analogue of starch is

A. cellulose
B. glycogen
C. sucrose
D. pentose

16) carbonmonosaccharides are called as

A. trioses
B. Tetrose
C. pentose
D. hexoses

17) Milk sugar is made from

A. maltose and fructose
B. sucrose and fructose
C. glucose and fructose
D. glucose and galactose

18) Formula for monosaccharide is

A. $(CH_2O)_n$
B. C_nH_{2n}
C. both A and B
D. none of above

19) Examples of polysaccharides includes

A. starch
B. cellulose
C. glycogen
D. all of above

20) Lactose is an example of

A. polysaccharides
B. monosaccharides
C. Disaccharides
D. Oligosaccharides

21) Cyclic monosaccharides contain number of carbon atoms

A. 5-6
B. 2-4
C. 1-3
D. 5-10

22) Condensation of two monosaccharides yields

A. disaccharides
B. polysaccharides
C. large amino acid chains
D. large β sheets

23) Units of carbohydrates which cannot be further hydrolyzed to simpler compounds are

A. disaccharides
B. polysaccharides
C. monosaccharides
D. Oligosaccharides

24) 6 carbon rings have suffix

A. glucose
B. fructose
C. pyranose
D. pentose

25) When carbon 5 attacks carbonyl carbon than fructose forms a

A. furanose
B. glucose
C. hexose
D. pentose

26) Linkage between two monosaccharides, is

A. ionic bond
B. covalent bond
C. hydrogen bond
D. glycosidic bond

27) All sugars end with

A. ase
B. ice
C. osi
D. ose

28) Starch is a glucose polymer in which glucopyranose units are bonded by

A. alpha linkages
B. beta linkages
C. gamma linkages
D. none of above

29) If OH group is to left of last stereocenter carbon than configuration is

A. D
B. L

C. A
D. Y

30) Reduction turns monosaccharides into

A. alcohols
B. Sugar alcohol
C. sesters
D. carboxylic acids

31331)

31) L and D are

A. epimers
B. enantiomers
C. both A and B
D. isomers

32) Smallest carbohydrate is

A. glyceraldehyde
B. glucose
C. fructose
D. galactose

33) Non-reducing disaccharides include

A. sucrose
B. trehalose
C. both A and B
D. maltose

34) On heating starches are

A. Insoluble in water
B. Soluble in water
C. becomes a regular geometrical shape
D. No effect of heat

35) Alternative name for table sugar is

A. ribose
B. sucrose
C. pentose
D. glucose

36) A monosaccharide switches from an open chain to a cyclic form through

A. hydroxylation
B. carbation
C. nucleophilic addition
D. hydrogenation

37) Maltose exists in

A. α configuration
B. β configuration
C. γ configuration
D. all of above

38) Characteristics combined to name monosaccharide carbohydrates include

A. number of carbon atoms
B. D or L configuration
C. placement of carbonyl group
D. all of above

39) ugar that makes up RNA is

A. ribose
B. deoxyribose
C. glucose
D. pentose

40) If carbonyl group is an aldehyde monosaccharide is

A. ketose

B. Tetrose
C. aldose
D. Maltose

41) For hydrolysis of starch, catalyst used is

A. glycosylase
B. amylase
C. polymerase
D. helicase

42) Carbohydrates are also termed as

A. crabs
B. carbs
C. cerbs
D. glycosides

43) Number of monosaccharides that condensed to form disaccharides structure is

A. 3
B. 4
C. 2
D. 1

44))) Brain gets energy from ketone bodies if availability of glucose is

A. constant
B. less
C. high
D. zero

45) Nature of anabolism reaction is

A. exergonic
B. endergonic
C. spontaneous
D. non spontaneous4

46) Number of water soluble molecules ketone bodies include are

A. 1
B. 2
C. 3
D. 4

47) n deamination, amino acid is converted into

A. aldol acid
B. keto acid
C. hydrochloric acid
D. carboxylic acid

48) Ketone bodies are made from

A. acetone
B. hydrochloric acid
C. acetyl-CoA
D. acetoacetic acid

49) Ketone bodies include

A. acetone
B. acetoacetic acid
C. beta-hydroxybutyric acid
D. all of above

50) In anabolism, cellular size

A. decreases
B. remains same
C. increases
D. becomes zero

51) Condition in which rate of synthesis of ketone bodies exceeds rate of utilization is called

A. ketonemia
B. anaemia

C. diabetes
D. color blindness

51) Ketone bodies cannot be used for energy by

A. liver
B. heart
C. brain
D. Eyes

52) Glycogen in liver is used for

A. plasma glucose
B. starch synthesis
C. amylose synthesis
D. Pentose hydrolysis

53) Ketone bodies are produced by

A. stomach
B. liver
C. kidneys
D. gallbladder

54) Process of breakdown of amino acids to α ketoacids is called

A. cisamination
B. amination
C. transamination
D. Racemization

55) Acetone is a ketone body which is a degradation breakdown product of

A. hydrochloric acid
B. acetic acid
C. citric acid
D. acetoacetic acid

56)Glycerol 1 phosphate reacts with acetyl CoA to produce

A. phosphoric acid
B. lactic acid
C. citric acid
D. phosphatidic acid

A57)Function of proteins is to

A. transport oxygen to hemoglobin
B. catalyze biochemical reactions
C. regulate reactions
D. all of above

58) Transamination is catalyzed by

A. transferases
B. aminotransferases
C. hydrogenases
D. dehydrogenases

59) Biosynthesis of complex molecules from simpler forms is called

A. anabolism
B. catabolism
C. metabolism
D. none of above

60) n SDS electrophoresis, proteins are separated on basis of

A. charge
B. mass
C. both A and B
D. structure

61) Interactions which holds and stabilizes sub units in 4° structure of proteins are

A. hydrophilic interactions
B. hydrogen bonding
C. hydrophobic interactions
D. Ionic bonding

62) Process of folding depends upon the

A. solvent
B. concentration of salts
C. pH
D. all of above

63) A technique which separates charged particles using electric field is

A. Hydrolysis
B. electrophoresis
C. protein synthesis
D. protein denaturing

64) Proteins that do not function but recover their activity upon folding are

A. denatured
B. folded
C. synthesized
D. hydrolyzed

65) At certain pH environments isoelectric point affects the

A. solubility of molecule
B. solubility of solvent
C. temperature
D. density of molecule

66) In peptide chain alpha helix is secured by

A. sulfur linkage
B. amide group
C. carboxyl group
D. hydrogen bonding

67) A process by which a protein structure assumes its functional shape or conformation is

A. denaturing
B. folding

C. synthesis
D. hydrolysis

68) Coiling and folding of polypeptide chain gives us

A. 1° structure
B. 2° structure
C. 3° structure
D. 4° structure

69) Amino acids interact with each other to produce a well-defined 3D structure known as

A. native state
B. denaturing
C. 1° structure of proteins
D. synthesis of proteins

70) Alpha-helix structure was discovered by

A.
B. Herman Branson
C. both A and B
D. Miller and Harley

71) PH at which amino acids does not migrate in an electric field in

A. isoelectric
B. electric point
C. field point
D. all of above

72) Linear sequence of amino acids forms

A. 1° structure
B. 2° structure
C. 3° structure
D. 4° structure

73). In beta pleated sheet, hydrogen bonding is between

A. sulfur linkage
B. amide group
C. polypeptide units of folded sheet
D. inside a single sheet

t74)) Type of bonding in disulfide bridges is

A. covalent bonding between R groups of Cysteine amino acids
B. ionic bonding
C. hydrogen bonding
D. polar interactions

75) Alternative name for spheroproteins is

A. Linear protein
B. Secondary proteins
C. Globular proteins
D. Beta structure

76) In glycolysis glucose is converted into

A. pyruvate
B. glycogen
C. sucrose
D. starch

77) Pentose phosphate pathway is also termed as

A. glycolysis
B. gluconeogenesis
C. phosphogluconate pathway
D. Citrate cycle

78) Gluconeogenesis is often associated with

A. ketosis
B. hexoses
C. pentose
D. Tetrose

79)Fermentation takes place in

A. absence of oxygen
B. presence of oxygen
C. presence of carbon
D. presence of nitrogen

80)Process of glycolysis occurs in

A. nucleus
B. cell wall
C. cell membrane
D. cytoplasm

81)Primary role of pentose phosphate pathway is

A. catabolic
B. anabolic
C. both A and B
D. Neutral

88)Gluconeogenesis is useful for

A. maintaining blood glucose level
B. avoiding low blood glucose level
C. both A and B
D. Maintain respiration rate

89)One glucose molecule can give number of pyruvate molecules

A. 3
B. 4
C. 2
D. 5

90) Amount of amino acid residues in proteins ranges from

A. 50-2000
B. 2000-4000
C. 4000-6000

D. 7000-10,000

91) Polar amino acids are usually found

A. on surface of proteins
B. inside core of proteins
C. at sides of proteins
D. can be present anywhere in proteins

92) Number of amino acids that have hydrophobic side chains are

A. 7
B. 8
C. 9
D. 10

93)Bond formed between two amino acid molecules is

A. peptide bond
B. sulfur linkage
C. ionic bond
D. coordinate covalent bond

94) At pH = 1, amino acid exists as

A. anion
B. cation
C. zwitterion
D. Neutral

95) At pH equal to pK_a amino acid solution exists as

A. cationic form
B. anionic form
C. zwitterion form
D. both A and C

96)Formation of dipeptide from two amino acids is accompanied by loss of

A. amine group
B. carboxyl group
C. sulfur
D. water molecule

97)Arginine, lysine and histidine have basic side changes at

A. Acidic pH
B. Basic pH
C. Neutral pH
D. High pressure

98) Symbol which indicates "Right" and is assigned clockwise in decreasing priority is

A. S
B. R
C. L
D. D

99)in hydrolysis of proteins energy is

A. absorbed
B. released
C. remains unchanged
D. none of above

100) Each amino acid unit in polypeptide chain is called

A. unit
B. residue
C. molecule
D. Element

101) One of few amino acids, cysteine is an amino acid that can be synthesized in human body and it contains

A. phosphoric acid
B. Sulfur
C. Phosphorous

D. Magnesium

102). Ionic state of amino acids depends upon

A. pH of solution
B. pH of solute
C. pH of solvent
D. Pressure

103)•D-amino acids are also non standard amino acids that occur naturally and include

–free D- serine, and D-aspartate in

A)Blood tissues

B) brain tissue

C)Skin tissues

D)All of these

104)Amino acids that can be found in proteins are

A. A- amino acids
B. B- amino acids
C. L- amino acids
D. D- amino acids

105). Amino acids in which -NH_3 is attached to carbon atom neighboringto -COOH group are

A. α amino acids
B. β amino acids
C. γ amino acids
D. none of above

106) D symbol indicates rotation in

A. anticlockwise rotation
B. clockwise rotation
C. spin up
D. spin down

107). Two symbols associated with absolute configuration

A. R
B. S
C. both A and B
D. L and D

108) The repeating units of proteins are
a) glucose units
b) amino acids
c) fatty acids
d) peptides

109). Amino acids are joined by
a) peptide bond
b) hydrogen bond
c) ionic bond
d) glycosidic bond

110). The primary structure of protein represents
a) Linear sequence of amino acids joined by peptide bond
b) 3-dimensional structure of protein
c) helical structure of protein
d) sub unit structure of protein

111). Peptide bond is
a) rigid with partial double bond character
b) planar, covalent
c) covalent

d) all of the above

112). Enzymes are

a) proteins

b) carbohydrates

c) nucleic acids

d) DNA molecule

113). The first protein sequenced by Frederick Sanger is

a) Haemoglobin

b) myoglobin

c) insulin

d) myosin

114). A dipeptide has

a) 2 amino acids and 1 peptide bond

b) 2 amino acids and 2 peptide bonds

c) 2 amino acids and 3 peptide bonds

d)2 amino acids and 4 peptide bonds

115). The most common secondary structure is

a) α-helix

b) β-pleated sheet

c) β-pleated sheet

d)all of these

116). Myoglobin is a

a) protein with primary structure

b) protein with secondary structure

c) protein with tertiary structure

d) protein with quaternery structure

117. Fibrous protein such as silk fibroin consists of polypeptide chains

arranged in
a) α-helix
b) β-pleated sheet
c) β-helix
d) none of these

118. α-helix has
a) 3.4 amino acid residues/turn
b) 3.6 amino acid residues/turn
c) 3.8 amino acid residues/turn
d) 3.0 amino acid residues/turn

119. Tertiary structure is maintained by
a) peptide bond
b) hydrogen bond
c) di-sulphide bond
d) all of the above

120). Haemoglobin has
a) primary structure
b) secondary structure
c) tertiary structure
d) quaternery structure

121). Disulphide bonds are formed between
a) cysteine residues that are close together
c) proline residues that are close together
d) histidine residues that are close together

122). The 3-D structure of protein can be determined by
a) Nuclear magnetic resonance
b) X-ray crystallography
c) both a and b

123) whicis is a type of dementia that causes problems with memory, thinking and behaviour.
A) Huntington's disease
B) Alzheimer's disease
C)BOTH OF THESE
d)none of these

123). Weak acids:
a) generally have pK_a values greater than 7.
b) generally have pK_a values between 2 and 10.
c) are another name for dilute acids.
d) cannot be used to buffer the pH of a solution.

124. Which of the following is not a sensible grouping of amino acids based on their polarity properties?
a) Arg, His, and Lys.
b) Phe, Trp, and Tyr.
c) Glu, Gly, and Val.
d) Asn, Gln, and Ser.

125). Which of the following pairs of amino acids would carry a positive charge on their side chain at pH 5.0
a) Asparagine & Glutamine.
b) Leucine& Glycine.
c) Histidine& Lysine.
d) Aspartate & Glutamate.

126)The peptide bond in proteins is:
a) only found between proline residues.
b) usually trans unless proline is the next amino acid.
c) is planar because of steric hinderance.
d) defines one of the angles used in the Ramachandran plot.

127). Hydrogen bonds in b-sheets are

a) more numerous than Van der Waals interactions.
b) not present at Phe residues.
c) roughly perpendicular to the polypeptide chain direction.
d) about 5Å in length.

128) Which of the following is considered to be a fibrous protein?
a) Prealbumin
b) Keratin.
c) Hemoglobin.
d) Protein G.

129). The major element of secondary structure in myoglobin and hemoglobin is:
a) the b-strand.
b) the a-helix.
c) the 310 helix.
d) All of the above.

130). Which of the following occurred when RNaseA refolded from a denatured state?
a) The primary structure of the protein was rearranged.
b) Most of the charged residues were found buried in the protein.
c) The entropy of the protein increased.
d) None of the above.

131) Which of the following is not an example of supersecondary structure?
a) Greek key
b) b-meander
c) bab unit
d) Type I reverse turn

132. Metal ion coordination

a) is a stabilizing force in some proteins.
b) generally denatures globular proteins.
c) occurs only in DNA and RNA.
d) causes tryptophan to absorb more visible light.

133). The Chime images on the web pages display structures determined by
a) X-Ray difraction.
b) NMR (nuclear magnetic resonance).
c) Both a) and b) are correct.
d) Neither a) nor b) are correct.

134). A tetrameric protein always has
a) four identical subunits.
b) four subunits.
c) four dissimilar subunits.
d) two each of two different subunits.

Q1)DefineGlycosidic bond?

Ans; Glycosidicbond of maltose is formed between the OH of carbon 1 and carbon 4 of 2nd glucose monomers.

Thereforer, it forms an alpha (1– 4) glycosidic bond.

Q2)DefineCarbohydrates?write the name of its major groups?

Ans;Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such
compounds on hydrolysis

• Many, but not all, carbohydrates have the empirical formula(CH2O)n; some also

contain nitrogen, phosphorus or sulfur

Classification

• Carbohydrates are classified into four major groups:

–Monosaccharides

–Oligosaccharides

• Disaccharides

–Polysaccharides

Q3)defineHomopolysaccharides and Heteropolysaccharides?

Ans;Homopolysaccharides (homoglycans): Polymers of same monosaccharide units e.g.

starch, glycogen, inulin, cellulose, dextrins, dextrans

2. Heteropolysaccharides (heteroglycans): Polymer of different monosaccharide units or

their derivatives e.g. Mucopolysaccharides (glycosaminoglycans).

Q4) Write the importance of Biomedical ?

Ans;Chief source of energy

• Constituents of compound lipids and conjugated proteins

• Lactose , the principal sugar of milk

• Degradation products utilized for synthesis of other substances such as fatty

acids, amino acids, cholesterol, etc

• Constituents of mucopolysaccharides which form the ground substance

ofmesenchymal tissues

• Certain carbohydrate derivatives are used as drugs like cardiac glycosides.

• Inherited deficiency of certain enzymes in metabolic pathways of different carbohydrates can

cause diseases e.g. galactosemia, glycogen storage diseases, lactose intolerance, etc

• Derangement of glucose metabolism is seen in Diabetes mellitus

Q5)Define Isomerism AND anomers?

Ans; The existence of two or more compounds having the same molecular formula but a

different arrangement of atoms within the molecule.

Isomeric forms of monosaccharides that differ only in their configuration about the

hemiacetal or hemiketal carbon atom are called anomers.

OR Isomeric forms of monosaccharides that differ only in their configuration about the

hemiacetal or hemiketal carbon atom are called anomers.

Q6)Define • Stereoisomers and enantiomers and diastereomers?

Ans; Compounds which are identical in composition and differ only in configuration (the

fixed spatial arrangement of atoms) are called stereoisomers

• Stereoisomers that are mirror images of each other are called enantiomers

• Pairs of stereoisomers that are not mirror images of each other are called diastereomers

Q7)define peptide bond?

Ans;Proteins are polymers of amino acids, with each amino acid residue joined to its neighbor

by a specific type of covalent bond termed the peptide bond.

Q8) Explain zwitterion?

Ans;The form of an amino acid that has both a positive and a negative charge is called a

zwitterion

• Amino acids in solution at neutral pH exist predominantly as dipolar ions (zwitterions)

• In this form, the amino group is protonated (–NH3+)

• and the carboxyl group is deprotonated

(–COO-)

• A zwitterion can act either as an acid (proton donor) or as a base (proton acceptor)

• Substances having this dual nature are amphoteric and are often called ampholytes

• The ionization state of an amino acid varies with pH

• At physiological pH, the carboxyl group is dissociated , forming the negatively

charged carboxylate ion (-COO

-

) and the amino group is protonated (NH3

+

)

• A simple monoamino monocarboxylic - amino acid, such as alanine, is a diprotic acid

when fully protonated—it has two groups, the-COOH group and the - NH_3^+ group, that can yield protons.

Q9)name This class of amino acids?

Ans; This class of amino acids includes

• serine,

threonine,

• cysteine,

• asparagine, and

• glutamine.

Q10)Name . Positively Charged (Basic) R Groups Amino acid ?

Ans; Positively Charged (Basic) R Groups

• The side chains of these amino acids are proton acceptors

• The amino acids in which the R groups have significant positive charge at pH 7.0 are:

– Lysine,

– Arginine,

-Histidine

Q11) Negatively Charged (Acidic) R Groups Amino acid?

Alanine Tyrosine

Arginine

Asparagine

Cysteine

Glutamate

Glutamine

Glycine

Proline

Serine

Q12)Explain The Structure of Proteins?

Ans;Structure of Proteins Four levels of protein structure are commonly defined

The Primary Structure

A description of all covalent bonds (mainly peptide bonds and disulfide bonds)

linking amino acid residues in a polypeptide chain is its primary structure

The most important element of primary structure is the sequence of amino acid

residues Secondary structure

Secondary structure refers to particularly stable arrangements of amino acid residues giving

rise to recurring structural patterns

Tertiary structure

Tertiary structure describes all aspects of the three-dimensional folding of a polypeptide

Quaternary structure

When a protein has two or more more polypeptide subunits, their arrangement in space is referred to as quaternary structure.

Q13)Write the name of Disease cause by misfoldind of protein?

Ans;Many conditions, including

• Alzheimer's disease; Alzheimer's disease due to deposition of amyloid-b protein cleaved from

amyloid precursor protein (APP).

• Type 2 diabetes,

• Huntington's disease and

• Parkinson's disease, arise from a common misfolding mechanism.

Q14)DEFINE INDUCE FIT MODLE?

ANS; The protein can discriminate among thousands of different molecules in its

environment and selectively bind only one or a few similar compounds.

• A given protein may have separate binding sites for several different ligands

• The binding of a protein and ligand is often coupled to a conformational change in

the protein that makes the binding site more complementary to the ligand, permitting

tighter binding

• The structural adaptation that occurs between protein and ligand is called induced fit.

• Reversible binding of ligands is essential for the function of gloubular proteins.

Q15)defineproteins?

Ans; Proteins are polymers composed of building blocks called amino acids, of which life on Earth uses just twenty. Molecular bonds determine the structures of amino acids and proteins. Peptide bonds link amino acids together in a chain; disulfide bridge bonds hold proteins together.

Q16)write the funcation of protein?

Ans; They are coded for by our genes and form the basis of living tissues. They also play a central role in biological processes. For example, proteins catalyse reactions in our bodies, transport molecules such as oxygen, keep us healthy as part of the immune system and transmit messages from cell to cell.

Q17)define Protein Denaturation?

A loss of three-dimensional structure sufficient to cause loss of function is called denaturation

(Electrostatic bonds,disulfide bonds, Specific hydrogen bonding, Hydrophobic bonds)

• The denatured state does not necessarily equate with complete unfolding of

the protein and randomization of conformation

• Under most conditions, denatured proteins exist in a set of partially folded stateMost proteins can be denatured by heat,.

Q18)Write a note on Protein Misfolding?

Ans; Protein Misfolding • Despite the many processes that assist in protein folding, misfolding does occur.

• Misfolding of proteins may occur spontaneously

• or caused by a mutation in a particular gene, producing an altered protein.

• In addition some apparently normal proteins can, after abnormal proteolytic cleavage,

take on a unique conformational state

• That leads to the formation of long fibrillar protein assemblies consisting of β

pleated sheets

• Many conditions, including

• Alzheimer's disease

• Type 2 diabetes,

• Huntington's disease and

• Parkinson's disease, arise from a common misfolding mechanism.

• In most cases, a soluble protein that is normally secreted from the cell is secreted

in a misfolded state

• and converted into an insoluble extracellular amyloid fiber.

• The diseases are collectively referred to as amyloidoses.

• Amyloidosis is a generic term that refers to the extracellular tissue deposition of

fibrils composed of low molecular weight of a variety of proteins..

• Amyloidoses may be

• organ specific or

• generalized (systemic).

• Primary systemic amyloidosis is caused by deposition of fibrils consisting of misfolded

immunoglobulin light chain

• Examples of organ-specific amyloidoses include:

• Alzheimer's disease due to deposition of amyloid-b protein cleaved from

amyloid precursor protein (APP).

• Islet amyloid polypeptide (IAPP) is commonly seen in diabetes mellitus type 2 and is

caused by deposition of amylin in pancreatic islets.

• It has been suggested that the disease-causing mechanism in Huntington's disease (and

the other polyglutamine disorders) is the ability of polyglutamine to undergo a

conformational change that can lead to the formation of very stable anti-parallel beta-

sheets; more specifically, amyloid structuresPhotomicrograph of amyloid plaques in a section of temporal cortex from a patient

with Alzheimer's disease

Alzheimer's is a type of dementia that causes problems with memory, thinking and behaviour. In

August 1994, at the age of 83, Reagan was diagnosed with Alzheimer's disease. Died in 2004.